

Introduction to computing

programming introduction

Computer science is about problem solving.

Type of knowledge:-

Declarative:- (سچے یا جھوٹ) (سچ یا جھوٹ)
 $\sqrt{36} = 6$

Imperative:- (سوالیہ کا جواب)

How?

$\sqrt{36} = ?$ we follow a process to do this → Algorithm

Case study: square root

How do you calculate the square root?

class Students

approach:-

$$\frac{36}{2} = \frac{18}{2} = \frac{9}{3} = \frac{3}{3} = 1$$

$$2 \times 3 = 6$$

- Start with "small num"
- divide
- result
- "pair" ^{must} same number
- get one element from each pair multiply together

assumption

Algorithm to find square root

now for 35 =

$$\frac{35}{5} = \frac{7}{7} = 1$$

×

35

×

identify problem

1 Step in problem solving:-

- Identify your assumption
 - Then write algorithm-
- My assumption is the num is > 0 .

jo bahar se input ay gi os ka baray ma ham na assumption dete ha

→ start guess - first 4 decimal place same

if guess "good enough?" good enough algorithm

return guess.

یہ گیسز کے قریب

improve algorithm

if not guess good "improve" guess -

g, num

g + n
0.33 → 0.34 → e.g
imp rate

start g with 0

$$\text{guess}^2 \approx \text{input}$$

$$6.0001^2 \approx 36$$

$$36.00001$$

$$1 \approx 36.0000$$

$$1.000$$

How to talk to computers?

- Programming languages (recipes)
by this you tell recipe to the computer.
- Interpreter (translator)
- Flow of control (working)
- Instruction (steps) modify data (ingredients)
in location (utensils)
جہاں آپ دیا
نے کھانا پکانا
دے رکھنے کے لیے
کڑی